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Editorial Office
Foundations of Physics

Dear Editors of *Foundations of Physics*,

I am pleased to submit my manuscript, "A Correlation-Production Reformulation of the Transfer-Entropy Second Law of Information Thermodynamics," for consideration as an Original Research Article in *Foundations of Physics*.

The manuscript revisits the transfer-entropy second law introduced by Ito by reformulating the inequality in terms of entropy change, mutual information, transfer entropy, reversed transfer entropy, and explicit system-bath correlations. The central purpose is not to propose a new model, but to clarify the informational decomposition underlying the second-law-type bound and to connect the transfer-entropy formulation with correlation-production approaches in stochastic thermodynamics. This perspective is intended to make transparent which terms represent directional information flow and which terms represent correlations accumulated between the system and bath.

I believe the manuscript is appropriate for *Foundations of Physics* because it addresses a foundational issue in information thermodynamics: how different informational representations of entropy production are related at the level of the second law. The work is concise and technical, but its aim is conceptual. In particular, the reformulation helps situate Ito's backward-transfer-entropy inequality within a broader correlation-production framework. This also relates to recent developments in which the same Ito2016 backward-transfer-entropy inequality has been used to sharpen thermodynamic uncertainty relation bounds under continuous quantum measurement and feedback.

The manuscript is original, has not been published elsewhere, and is not under consideration by another journal. No conflicts of interest are declared. The work did not receive external funding.

Thank you for considering this submission. I would be grateful for the opportunity to have the manuscript evaluated by *Foundations of Physics*.

Sincerely,

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